

Chamali Abeysekara

(+94) 74-366-9386 indummaa2003@gmail.com linkedin.com/in/chamali-abeysekara
github.com/Chamali27



PROFILE

Applied Data Science undergraduate with a strong focus on data analytics, visualization, and business insights. Experienced in SQL Server, Power BI, Python, and R, with hands-on involvement in 10+ real-world projects covering data cleaning, analysis, and dashboard development. Strong communication skills with the ability to present insights clearly and translate complex data into actionable information for data-driven decision-making.

EDUCATION

Bachelor of Science in Applied Data Science Communication

Jan 2024 - Present

General Sir John Kotelawela Defence University

Advanced Level - Accounting (A), Economics (B), Business Statistics (C)

Completed 2023

Musaeus College

Ordinary Level - 5A, 3B, 1C

2009-2019

Vidura College

TECHNICAL SKILLS

Programming Languages: Python, SQL, R

BI & Visualization: Power BI, DAX, Microsoft Excel

Tools & Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, NLTK

Databases: MYSQL, SQL Server

Coursework: Machine Learning, Data Mining, Database Design

PROJECTS

An Analysis of U.S. Flight Arrival Performance | SQL Server, Power BI

2026

- Analyzed 102.84 million domestic flight records (2010 - 2025) from the BTS TranStats OT Delay dataset to assess delays, cancellations, and diversions across airlines and airports.

Sri Lanka Tourist Arrivals Analysis | Power BI, MS Excel

2026

- Analyzed 7 million tourist arrivals (2020 - 2025) using official SLTDA government data, examining recovery trends, source markets, and seasonal patterns across 234 countries.

Policy AI System | NLP, Generative AI

2026

- Built an AI application to extract and summarize key insights from policy documents using NLP (NLTK, TF-IDF) and large language models..

USD/LKR Exchange Rate Forecasting | Python, Machine Learning (Presented-ADScAI 26)

2026

- Built a CRISP-DM-based ML model to forecast USD/LKR rates using 3,300+ observations (2010–2026), applying feature engineering and time-series analysis; achieved $R^2 > 0.90$ with a 30-day recursive forecasting system.

UK Data Analytics Project | SQL Server, Power BI

2025

- Analyzed UK visa (2010 - 2024) and Manchester residential EPC datasets (2014–2024) to identify trends in migration patterns, energy efficiency, costs, and CO₂ emissions.

Global Model for EV Adoption - Environmental & Economic Impacts | Data Analysis, Modelling

2025

- Developed a global model to quantify environmental and economic benefits of EV adoption versus ICE vehicles, projecting CO₂ savings of 1.95 tons per vehicle per year.

- Prediction of Road Accident Severity** | *Python, Scikit-learn, Pandas, Matplotlib, Seaborn* 2025
- Built and compared Logistic Regression and Random Forest models on a 7.7M-record US accidents dataset to predict accident severity, achieving 80%+ accuracy with key insights on environmental risk factors.
- Exploring Consumer Behavior and Patient Outcomes** | *Association Rule Mining, Logistic Regression* 2025
- Applied Association Rule Mining on retail transaction data and Logistic Regression on healthcare patient data for predictive analytics.

CERTIFICATES

- Microsoft Certified: Azure Data Fundamentals (DP-900)** | *Microsoft* April 2026
- MATLAB Onramp - 100% Completed** | *MathWorks (Training Services)* 2024
- Certificate in Computer Applications - Distinction Pass** | *University of Colombo* May 2023 - Nov 2023
- 2nd Class Certificate of Merit** | *Musaeus College* 2022

ACHIEVEMENTS

- Research Paper Presenter - ADScAI 2026 - Applied Data Science & AI Conference, University of Moratuwa** 2026
- Presented “Predicting USD/LKR Exchange Rate: A CRISP-DM Machine Learning Approach”.
- Workshop Participant - ADScAI 2026 - Applied Data Science & AI Conference, University of Moratuwa** 2026
- Participated in Event-Driven AI on AWS (Emojot) and AI Agents (Cut+Dry), gaining exposure to AI, data science, and cloud applications.
- Data Odyssey 2025 - Champions (1st Place)** | *DatAIInspire KDU* 2025
- Designed and developed *Cart Connect*, an intelligent shopping trolley system using Association Rule Mining for real-time product recommendations, item detection, and smart retail navigation.
- Certificate of Recognition - Dancing Performances** | *General Sir John Kotelawala Defence University* 2024
- Recognized for commitment and dedication in university dancing performances at Colours Night 2024.
- Participant - Data Odyssey Inter-University Exhibition Competition** | *DatAIInspire KDU* 2024
- Proposed an AI-based mobile application for precision agriculture in Sri Lanka to improve farming efficiency and sustainability through data-driven recommendations and machine learning concepts.

LEADERSHIP & EXTRACURRICULAR

- Vice President, DatAIInspire KDU** Jan 2025 - Jan 2026
- Led KDU AI & Data Science Club, organizing workshops, hackathons, and the university-wide Data Odyssey competition while guiding a committee to foster innovation and analytical thinking across faculties.
- Assistant Secretary, Dancing Club (KDU)** 2025
- Assisted in coordinating dance club activities and event organization, while actively performing in university and inter-faculty events, contributing to teamwork and creative expression.
- Batch Representative - BSc in Applied Data Science Communication, Semester 03** 2024
- Facilitated communication between students and academic staff, coordinating batch-level academic and extracurricular activities.
- Member, KDU Toastmasters Club (District 82)** 2024 - Present
- Developing public speaking and communication skills through structured speeches and evaluations, while strengthening leadership, goal setting, and completing Dynamic Leadership Level 1.

REFERENCES

Ms. Chamodhi Wickramasinghe

Lecturer, General Sir John Kotelawala Defence University
+94713612792
wickramasinghe.bct@kdu.ac.lk

Dr. Charith Silva

Senior Lecturer, General Sir John Kotelawala Defence University
+447920206338
charithsilva@kdu.ac.lk